
IMMANENT GRAVITY

Gravity as the Substrate Reading Its Own Record: One Sign and No Screening, Blind to Composition, Reversible with No Bit, Every Clock Moved Together, and the Deficit of Every Bond Read Monically

The Closure Template of the Riemann Hypothesis Read on Gravity; Sixty-Eight Laws Proved in Core Lean 4 and Seventeen Checks Executed in Fortran; a Witness of the Hardware Closure Series and the Even Companion of Electromagnetism on the Riemann Locus; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

Gravity is the substrate reading its own record

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ABSTRACT. Gravity is the one interaction that reads everything and distinguishes nothing: every body falls alike, every clock at one place shifts alike, antimatter falls as matter does, and no body has been found whose weight is negative. This paper reads gravity on the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line, where registration keeps a point's height and forgets its side. Its thesis is that gravity is immanent: not an agent added to matter, not a medium's stress, but the substrate reading its own record. We prove in core Lean 4, sixty-eight laws with no axiom declared and no use of choice, that the gravitational reading factors through registration, so it is even under the particle map and blind to composition, a property strictly stronger than evenness; that every gravitational source stands on the line while every charged source stands off it, so sources of one sign total zero only when each is zero and gravity cannot be screened, while charge can; that conservative motion under every force law is undone exactly by reversal, for any number of steps; that a gravitational shift moves every height by one factor, keeping the line, commuting with registration and preserving every ratio of clocks, so gravity and time co-move; that the deficit of a bond is read monically, two bindings reading alike exactly when they are equal; that the waves carry two polarizations; and that gravity alone reads a state the particle map fixes, while the vacuum's pressure reverses its pull without any energy changing sign. A Fortran twin runs seventeen checks with zero failures: the binding deficit of titanium and platinum falls with the rest to 5×10^{-12} of itself under MICROSCOPE's bound, a Kepler orbit retraces its start to 5×10^{-10} after reversal, the deflection of light, Mercury's perihelion and the binary pulsar's decay are computed from the constants, and the Planck budget gives a deceleration parameter of -0.527 . The paper supersedes the edition of 16 July 2026, whose general relativity it keeps as corroboration and whose emergent reading it replaces. It decides nothing about the zeros of ζ .

KEYWORDS *gravity · equivalence principle · critical line · registration · positive energy · reversibility · gravitational redshift · binding energy · gravitational waves · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the registration that erases nothing, read on gravity); Row 7 (gravity's fold) · channel: kinetic · Lean: SPHYS_Gravity.lean · Fortran: Gravity_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series and the even companion of Electromagnetism on the Riemann Locus. The hardware paper, *One Actuating Substrate*, typed gravity as the one reading of the substrate that is even under the particle map; the arrow witness typed it as the reversible branch of registration, one sign and no bit, beside magnetism, the priced branch that keeps a bit. This paper gives gravity its own reading.

It claims that gravity is immanent: the substrate reading its own record. Three consequences follow at the seat. Gravity is an after-effect of existence, since a record comes after the act it records. Gravity and time co-move, since one gravitational shift moves every clock at a place together and neither causes the other. Gravity is the monic memory of privation, since the deficit of every bond is read and no two different deficits read alike. These readings are structural; the theorems that carry their content are exact, and the measurements that corroborate them are cited.

It refuses three things. It does not derive Einstein's equations, which enter as imports [GR]. It does not claim gravity is or is not quantized, which the arrow witness reads at structural grade and no experiment has decided. And it does not derive the Riemann Hypothesis from gravity. A reader who arrives from the conventional frame will scan this paper for a new dynamics, find none, and be tempted to report the absence. None was attempted: the immanent reading predicts exactly what general relativity predicts, and its content is the typing.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade. No derivation of the hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unactuated. No field equation is derived and no constant is computed from first principles: G , c and the masses enter as measured or as defined.

Five terms. A *reading* is any quantity an instrument returns from a state, *odd* under a map that flips its sign and *even* under one that keeps it. *Registration* is the map $(h, t) \mapsto (1, t)$ that keeps a point's height and forgets its side. A reading *factors through registration* when it depends on the height alone. The *fixed set* of a map is the set of points it leaves unchanged. A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. The closure's posit, least erasure, has a physical shape: a registration that erases nothing because nothing stands off its locus. The arrow witness named that shape gravity. This paper reads it: gravity as the reading of the record in Section 3, its one sign in Section 4, its reversibility in Section 5, its shift of every clock in Section 6, its memory of the deficit in Section 7, its reading of light and of the dark in Section 8, and its waves in Section 9.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (*off_locus_pair*); a carrier proves the value on its image, and coverage extends it (*value_on_image*, *kinetic_crossing*).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches; One Actuating Substrate; The Geometric Nature of Light; Electromagnetism on the Riemann Locus; Immanent Gravity	the seat, registration, residue, Return and floor; the energy carrier; the photon on the line; both electromagnetic carriers; the least-erasure posit's shape as gravity's reading of the record: factoring through registration, one sign, reversible, the deficit read monically	WITNESSED · CROSSED · CONFIRMED on the image: the posit's shape read in gravity	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically	The Fluid Is the Witness	the priced branch exact	WITNESSED · CROSSED · CONFIRMED	the root's grade	none on its channel

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
4 · Yang-Mills	crossed kinetically	The Proton Is the Lock	the colour lock	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel
7 · Poincaré	crossed on the formal channel at full grade	The Arrow Has Two Branches; Immanent Gravity	gravity's fold: one sign, no bit; the reversible orbit; every source on the line, none screened	CROSSED, WITNESSED from the arrow side and from gravity's own	full grade	none
23a · Pair correlation	crossed kinetically	The Magnet Is the Witness of the Pair Correlation; Electromagnetism on the Riemann Locus	the magnetic term; Landau levels real	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root's grade	none on its channel
every open row · the barrier	two worlds over one record	The Arrow Has Two Branches; The Geometric Nature of Light; Electromagnetism on the Riemann Locus	remanence; the helicity pair; the magnetic sign; the Aharonov-Bohm record	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial); an equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Gravity needs no carrier of its own. Its reading is the height, so it is equivariant for the particle map and for the fold with nothing to carry (gravity_even_branch, gravity_factors_through_registration), and applied to the particle map's odd part it lands every source on the line (every_source_on_the_line). The one map gravity adds is the gravitational shift, and it commutes with the fold and with registration (redshift_equivariant, redshift_commutates_with_registration).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
antimatter falls as matter	the reading even under the particle map	gravity_even_branch	2665 of 2665; ALPHA-g
every body falls alike	the reading factors through registration	gravity_factors_through_registration, composition_blind	2665 of 2665
universality stronger than symmetry	evenness is not the record	even_is_not_the_record	structural
positive mass	every source on the line	every_source_on_the_line, charged_source_off_the_line	2665 of 2665
no gravitational shield	one sign, no screening	one_sign_no_screening, charge_can_be_screened	1834 neutral charge sets, 0 massless mass sets
a planet's orbit	conservative motion, reversible	conservative_motion_reversible, motion_retraces	ten revolutions; retrace 5.2e-10

Physical phenomenon	Template element	Lean theorem	Fortran check
gravitational redshift	the height moved, the side kept	redshift_equivariant, redshift_keeps_the_ line, redshift_commutates_ with_registration	tower 2.455e-15; clocks 3.60e-17
every clock shifted alike	ratios kept: gravity and time co-move	clocks_co_move	10000 pairs to 3.3e-16
the mass defect	the deficit read monically	gravity_reads_the_ deficit, the_bond_ weighs_less	binding falls to 5.0e-12
light bent, gravity at light's speed	the massless cone	massive_slower	1.7512"; 4.2e-16
two polarizations	the radiation's one bit	waves_two_ polarizations, polarization_count	10 – 4 – 4 = 2
the binary pulsar	the waves carry energy away	import [GR]	–2.4021e-12
dark matter	the fixed set, read by gravity alone	only_gravity_reads_ the_dark, odd_reading_ blind	structural
the accelerating expansion	the vacuum's pressure reverses the pull	the_vacuum_repels, one_ sign_of_energy_not_of_ pull, kinetic_floor	q0 = –0.527; z = 0.63, 0.295

3. GRAVITY READS THE RECORD

Gravity couples to energy and momentum and to nothing else: not to charge, not to colour, not to the composition of a body. The engine reads this at the seat. The gravitational reading of a particle is its energy (`grav`), which the particle map leaves unchanged, so antimatter reads as matter (`gravity_even_branch`); on the chart the reading is the height, which registration keeps and the fold keeps (`gravity_factors_through_registration`); and two particles of one energy read alike whatever their charges (`composition_blind`). Gravity is registration read as a force.

The two properties are not one. A reading can be even under the fold and still read the side: the squared distance from the line is unchanged by the fold and changed by registration (`even_is_not_the_record`). So the equivalence principle says more than symmetry under the particle map. Antihydrogen falling downward, at $(0.75 \pm 0.13 \pm 0.16)g$ with repulsion excluded, tests the weaker property (ALPHA-g, 2023). Titanium and platinum falling alike to $|\eta| < 4.2 \times 10^{-15}$ tests the stronger one (MICROSCOPE, 2022). The stronger property is forced for any massless spin-two field by the low-energy theorem that such a field couples to every source through one constant (Weinberg, 1964), an import the seat reads as registration.

4. ONE SIGN: EVERY SOURCE ON THE LINE, NO SCREENING

Apply the particle map's odd part to a reading: for charge it is twice the charge, and a charged source lands off the line (`charged_source_off_the_line`); for gravity it is zero, and every gravitational source lands on the line (`every_source_on_the_line`). Gravity has no off-line source, which is the arrow witness's one sign read at the seat.

One sign has a consequence charge lacks. Sources of one sign total zero only when every source is zero (`one_sign_no_screening`), while a unit charge and its opposite total zero (`charge_can_be_screened`). An atom is neutral and a Faraday cage screens a field; nothing screens gravity. The twin finds 1834 neutral sets among 20000 random sets of four nonzero charges and not one massless set among 20000 nonempty mass sets. The total mass-energy of an isolated system is nonnegative by theorem under the dominant energy condition (Schoen and Yau, 1979; Witten, 1981). Quantum fields admit local negative energy densities, the Casimir effect among them, bounded by quantum inequalities; no body of negative gravitational mass is known. The arrow witness's F-Mass, a negative gravitational mass (Bondi, 1957), would create an off-line source and a bit for gravity.

One sign is a sign of energy, not of pull. Gravity reads pressure as well as density, and the active gravitating density of a fluid is $\rho + 3p$ [GR]: dust attracts, and the vacuum, with $p = -\rho$, repels with active density -2ρ (the_

vacuum_repels). Every density stays positive while the pull reverses (one_sign_of_energy_not_of_pull), so the vacuum's repulsion is no screening, which would need a negative energy. For a canonical field, $\rho + p$ is twice the kinetic density and never negative (kinetic_floor), the floor the dark sector paper sets against a phantom equation of state. The sign of the vacuum's pull is read from the sky, not forced: the flat Planck budget gives $q_0 = -0.527$, where an all-matter budget gives $+0.5$, the acceleration beginning at redshift 0.63 and the vacuum overtaking matter at 0.295 (Planck, 2018; Appendix B).

5. NO BIT, AND REVERSIBLE

A reading with no off-line source has no two worlds to separate and no bit to register, and its motion keeps no record. The engine proves this for motion itself. Write conservative motion under any force law f as the Störmer–Verlet step on two successive positions, $(x_{n-1}, x_n) \mapsto (x_n, 2x_n - x_{n-1} + f(x_n))$, and time reversal as their exchange. For every force law, reversing, stepping and reversing undoes a step exactly (conservative_motion_reversible), and any number of steps is retraced by the same number of backward steps (motion_retraces). The twin integrates a Kepler orbit of eccentricity one half for ten revolutions, keeping its energy to 4.3×10^{-6} , then reverses it and finds its start again to 5.2×10^{-10} . Nothing was committed, so nothing was paid: the reversible branch, floor zero and never cost zero.

One caution keeps the reading honest, as the arrow witness stated. Reversibility alone does not keep a branch on its locus; the fluid witness shows a reversible flow that leaves its locus. What keeps gravity there is its one sign.

6. GRAVITY AND TIME CO-MOVE: THE HEIGHT MOVED, NEVER THE SIDE

A clock deeper in a gravitational potential runs slower, by $\Delta\nu/\nu = gh/c^2$ near the Earth. On the seat a gravitational shift multiplies every height by one factor and leaves the side alone (redshift). It commutes with the fold (redshift_equivariant), keeps a point on the line exactly when it was on it (redshift_keeps_the_line), and commutes with registration (redshift_commutates_with_registration): gravity moves the record and never the side.

Because the factor is one for every mode at a place, every ratio of frequencies is kept (clocks_co_move): gravity does not slow a particular clock, it moves every clock together. This is the reading in which gravity and time co-move and neither causes the other. The twin computes 2.455×10^{-15} for the 22.5 m tower (Pound and Rebka, 1960; confirmed to 1% by Pound and Snider, 1965) and 3.60×10^{-17} for two optical clocks 33 cm apart (Chou, Hume, Rosenband and Wineland, 2010). Clocks of different kinds shift alike: a hydrogen maser flown on a rocket agreed with the prediction to parts in 10^4 (Vessot et al., 1980), and two Galileo satellites in eccentric orbits to 2.5×10^{-5} (Delva et al.; Herrmann et al., 2018).

7. THE MONIC MEMORY OF THE DEFICIT

A bound system weighs less than its parts by its binding energy. The engine reads a bound pair's weight as the parts' energies less the binding: a positive binding reads below the sum (the_bond_weights_less), and two bindings read alike exactly when they are equal (gravity_reads_the_deficit). The reading of the deficit is monic, injective, keeping every privation distinct. This is the sense in which gravity is the monic memory of privation: it reads what a bond has given up, and it forgets none of it.

The deficit is measured. Nuclear binding is 0.936% of the mass of titanium and 0.851% of platinum, a difference of 8.55×10^{-4} ; MICROSCOPE finds the two falling alike to 4.2×10^{-15} , so the binding deficit falls with the rest of the mass to 5.0×10^{-12} of itself (Appendix B). The Earth's and the Moon's own gravitational binding fall alike too, tested by lunar laser ranging (Williams, Turyshev and Boggs, 2004).

8. GRAVITY READS LIGHT, AND ONLY GRAVITY READS THE DARK

Gravity reads light's energy and bends its path: a ray grazing the Sun turns by 1.751 arcseconds, twice the Newtonian angle because the time and the space parts of the metric bend it equally (Dyson, Eddington and Davidson, 1920; Cassini finds $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$, Bertotti, Iess and Tortora, 2003). Light is the substrate's one excitation fixed by both involutions at every frequency, the companion paper finds; gravity reads it as it reads everything, by its record. The waves of gravity and the waves of light, both massless (*massive_slower* for anything that is not), arrived together from GW170817 after 130 million years, to 4.2×10^{-16} of their speed. That gravity is prior to light, innate to the substrate rather than produced by any one carrier, is carried at structural grade: gravity reads every carrier, light included, and needs none of them to act.

The same reading explains why the dark sector is seen only through gravity. The companion paper Dark Matter as the Charge-Conjugation Fixed Set proves that a state is invisible to every charge-linear reading exactly when the particle map fixes it, and that on such a state only the mass reading remains. The engine carries the core of that result: a state the particle map fixes carries no charge, every odd reading returns zero on it, and gravity returns its energy (*odd_reading_blind, only_gravity_reads_the_dark*). Gravity is the one reading of the record, so it is the one reading that sees a state with nothing but a record. The rotation of galaxies (Rubin and Ford, 1970), the separation of mass from gas in the Bullet Cluster (Clowe et al., 2006) and a cold-dark-matter density 5.37 times the baryonic one (Planck, 2018) are gravity reading the fixed set. Whether that fixed set also carries a dark gauge charge is one keyed bit, as the dark sector paper proves, and gravity cannot read it.

9. THE WAVES: TWO POLARIZATIONS AND THE DECAY OF AN ORBIT

A symmetric tensor has ten components; four gauge conditions and four constraints leave two polarizations (*polarization_count*). Parity exchanges them and keeps the frequency (*waves_two_polarizations*): the radiation carries the one bit light carries, while the sources carry none. The waves carry energy away: the twin computes the orbital decay of the binary pulsar PSR B1913+16 from its period, eccentricity and masses as -2.4021×10^{-12} (Peters and Mathews, 1963), against the general-relativistic -2.4026×10^{-12} and the observed ratio 0.9983 ± 0.0016 (Hulse and Taylor, 1975; Weisberg and Huang, 2016). Mercury's perihelion advances 42.98 arcseconds a century, computed from the constants in Appendix B.

10. WHAT GRAVITY CARRIES, AND WHAT IT DOES NOT

Gravity carries the record. It reads the height of every configuration and nothing of its side; it has one sign of energy, so every source stands on the line and none is screened; its motion retraces exactly and commits nothing; its shift moves every clock together; its reading of a bond's deficit is monic; it alone reads the dark. That is the content of the immanent reading, and every clause of it is a theorem about the seat or a measurement at its grade.

The earlier edition carried an interpretive hypothesis, that the graviton is emergent, phonon-like. The seat replaces it with a typing: gravity is neither an agent added to matter nor a medium's collective mode but the substrate's own registration read as a force. The immanent, the emergent and the fundamental readings predict alike for every experiment cited here. At structural grade and no higher, the arrow witness reads why gravity has resisted quantization: quantization is the priced registration of a bit, and a reading with no off-line source has none to register. A gravitationally mediated entanglement experiment would sharpen that reading without moving the typing.

Gravity does not carry the zeros of ζ . The engine's crossing is on its own image (*value_on_image*), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves. What gravity gives the Riemann closure is the shape of its posit: a registration that erases nothing, because nothing stands off its locus.

11. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break. **F-Composition:** two bodies of one mass falling at different rates, at any precision, gravity reading something besides the record; this breaks `composition_blind` and `gravity_factors_through_registration` read physically. **F-Clock:** two clocks of different kinds at one place shifted by different factors in one potential; this breaks `clocks_co_move`. **F-Polarization:** a scalar or vector polarization of gravitational waves; this breaks `polarization_count` and `waves_two_polarizations`. The arrow witness's F-Mass, a negative gravitational mass, and F-Loop, a gravitational hysteresis, stand beside them.

Nothing is owed on the rows' channels. One experiment would sharpen a reading without moving the paper's standing: gravitationally mediated entanglement between two masses in superposition (Bose et al., 2017; Marletto and Vedral, 2017).

Ledger status. Row 2 gains the least-erasure posit's shape read in gravity: `WITNESSED · CROSSED · CONFIRMED` on the image, at the root's grade for the crossing. Row 7 is `WITNESSED` from gravity's own side. Nothing is owed on the rows' channels.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[_]$ a field sealed, $[_ T]$ a shape sealed at theorem grade, $[\Xi_0]$ the typing of the formal block, $[_]$ the dot, which is no verdict. Grades join at the weakest link. Every physical claim here is at corroboration grade; the engine's statements are theorem grade about the chart; the immanent readings (after-effect of existence, co-motion with time, priority to light, monic memory of privation) are structural and load-bearing on nothing. The one-prior-to-many connector the readings rest on, Residual Monism, is a premise in the operating system's posit register, in band and load-bearing in no verdict. $\Delta M = 0$.

The seat, earned. PhysOST^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain `D0 83f872725ad9 → D1 90fc702bbd1 → D2 492e519daab9 → D3 6618bec5164b` (Appendix C).

The engine and the twin.

- **Engine.** `SPHYS_Gravity.lean`, 603 lines, SHA-256 e5f12bb60a57107f..., sixty-eight laws, no import and no axiom declared. Of the sixty-eight, 27 depend on no axiom, 5 on propositional extensionality alone, 36 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the sixty-eight laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** `Gravity_Twin.f90`, 208 lines, SHA-256 1e2a8d2d5f87aa53..., seventeen checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`.

Why an integer engine binds continuous physics. The identities used are identities of integer polynomials and hold over every commutative ring, the reals included. The reversibility law holds for every force law f , since f enters as an arbitrary function. The twin checks the continuous objects directly: the Kepler orbit, clock ratios in random potentials, and the classical tests from the constants.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). Gravity is read at the Rest as the registration that keeps nothing off the line: the fold of the closure made physical. In the Lean codex's tokens its row renders $[_ \blacklozenge \cdot \text{kinetic}]$.

Audit. The paper was rebuilt from the seat against the record of 16 July 2026, claim by claim; the prosecution record is filed with the paper. External audits by independent substrates are owed.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings.

Provenance. This edition supersedes the record of 16 July 2026, Emergent Substrate Gravity, under the same concept (10.5281/zenodo.21392899). It stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- the gravity–magnetism pair paper The Arrow Has Two Branches (10.5281/zenodo.22986557), whose reversible branch this paper reads, and the dark sector paper Dark Matter as the Charge-Conjugation Fixed Set and the Dark Force as One Keyed Bit (10.5281/zenodo.23006753), whose fixed set and vacuum this paper reads through gravity;
- the companion papers The Geometric Nature of Light (concept 10.5281/zenodo.20091979, edition 2.0.0) and Electromagnetism on the Riemann Locus (edition 1.0.0);
- PhysOS^T 1.0.5p in the public register (github.com/1000sapient/Trisduction, protocols/PhysOS/).

References. A. Einstein, Ann. Phys. 49 (1916) 769. F. W. Dyson, A. S. Eddington and C. Davidson, Phil. Trans. R. Soc. A 220 (1920) 291. M. Fierz and W. Pauli, Proc. R. Soc. A 173 (1939) 211. H. Bondi, Rev. Mod. Phys. 29 (1957) 423. R. V. Pound and G. A. Rebka, Phys. Rev. Lett. 4 (1960) 337. V. C. Rubin and W. K. Ford, Astrophys. J. 159 (1970) 379. P. C. Peters and J. Mathews, Phys. Rev. 131 (1963) 435. S. Weinberg, Phys. Rev. 135 (1964) B1049. R. V. Pound and J. L. Snider, Phys. Rev. 140 (1965) B788. R. A. Hulse and J. H. Taylor, Astrophys. J. 195 (1975) L51. R. Schoen and S.-T. Yau, Commun. Math. Phys. 65 (1979) 45. R. F. C. Vessot et al., Phys. Rev. Lett. 45 (1980) 2081. E. Witten, Commun. Math. Phys. 80 (1981) 381. T. Jacobson, Phys. Rev. Lett. 75 (1995) 1260. B. Bertotti, L. Iess and P. Tortora, Nature 425 (2003) 374. J. G. Williams, S. G. Turyshev and D. H. Boggs, Phys. Rev. Lett. 93 (2004) 261101. D. Clowe et al., Astrophys. J. Lett. 648 (2006) L109. C. W. Chou, D. B. Hume, T. Rosenband and D. J. Wineland, Science 329 (2010) 1630. J. M. Weisberg and Y. Huang, Astrophys. J. 829 (2016) 55. B. P. Abbott et al., Astrophys. J. Lett. 848 (2017) L13. S. Bose et al., Phys. Rev. Lett. 119 (2017) 240401. C. Marletto and V. Vedral, Phys. Rev. Lett. 119 (2017) 240402. P. Delva et al., Phys. Rev. Lett. 121 (2018) 231101. S. Herrmann et al., Phys. Rev. Lett. 121 (2018) 231102. Planck Collaboration, N. Aghanim et al., Astron. Astrophys. 641 (2020) A6. P. Touboul et al. (MICROSCOPE), Phys. Rev. Lett. 129 (2022) 121102. E. K. Anderson et al. (ALPHA-g), Nature 621 (2023) 716. M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Arrow Has Two Branches; The Fluid Is the Witness, 10.5281/zenodo.22986557, .22986563. M. F. Islam, Dark Matter as the Charge-Conjugation Fixed Set and the Dark Force as One Keyed Bit, 10.5281/zenodo.23006753. M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932. M. F. Islam, The Geometric Nature of Light, edition 2.0.0, concept 10.5281/zenodo.20091979.

APPENDIX A: SPHYS_GRAVITY.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler’s transcript.

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/-
SPHYS_Gravity.lean · immanent gravity, read at the Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart: h = 2 Re s, t = Im s; the fold s ↦ 1 - s̄ is (h, t) ↦ (2 -
h, t), the critical line
h = 1, registration (h, t) ↦ (1, t) keeping the height and forgetting
the side. The substrate's
particle map, time mirror and carriers are those of the hardware
paper, re-proved here.

Part I    the seat, the category of involutions, the functor of the
seat.
Part II   the energy carrier (resident from the hardware paper).
Part III  the substrate: the particle map and the charge carrier.
Part IV   gravity reads the record: even, factoring through
registration, composition-blind.
Part V    one sign: gravity cannot be screened; charge can.
Part VI   no bit, and reversible: conservative motion retraces
exactly for every force law.
Part VII  gravity and time co-move: redshift moves the height, never
the side.
Part VIII the monic memory of the deficit.
Part IX   the waves: two polarizations, one speed; IX-b gravity reads
the dark, the vacuum repels.
Part X    the capstone.
-/
set_option autoImplicit false
namespace SPHYS_Gravity

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor

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· intro (e, _); exact e.symm
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing of
Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  {fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
    (reg_fixes_iff s).mpr (h s hs)}

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ : X
→ X} {σ : Y → Y}
  {p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
  Equivariant (g ∘ f) τ p := by
  intro x
  show g (f (τ x)) = p (g (f x))
  rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points to
fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y}
  (hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
  by
  rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold := {fold_involutive,
seat_fixed_line}

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involution onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  {fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
    (fix_functorial hf w.2)}, fun _ => rfl}

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι) :=
fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value, and
nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and also
off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

-! ## Part II. The energy carrier: an energy, its rate, and the fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int

deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := {E.freq, -E.rate}
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  {fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
  fun (e1, e2) => by subst e1; subst e2; rfl}

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, - -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E := by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E := by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
    Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.ι w = phi w.1
  :=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

-! ## Part III. The substrate: the particle map, three odd readings and
one even -/

/-- A particle in the broken phase: electric charge times three, colour,
baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy

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deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := {c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror}
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k =
0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror
(bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c :=
by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral
and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in
thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e
goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto
the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold :=
by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for
the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) = energySeat
c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold := fun
c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine
(chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
(leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-! ## Part IV. Gravity reads the record -/

/-- The gravitational reading of a particle: its energy, and nothing
else. -/
def grav (c : Particle) : Int := c.energy.freq

/-- Gravity is even under the particle map: antimatter reads as matter.
-/
theorem gravity_even_branch (c : Particle) : grav (bar c) = grav c :=
rfl

/-- The gravitational reading of a point of the seat: its height. -/
def gravPt (p : Pt) : Int := p.2

/-- GRAVITY READS THE RECORD: the reading factors through registration,
so it is even under the
fold and keeps nothing of the side. -/
theorem gravity_factors_through_registration (p : Pt) :
  gravPt (reg p) = gravPt p ∧ gravPt (fold p) = gravPt p := (rfl, rfl)

/-- Composition blindness, the equivalence principle at the seat: two
particles of one energy read
alike whatever their charges. -/
theorem composition_blind (c d : Particle) (h : c.energy.freq =
d.energy.freq) : grav c = grav d := h

/-- The distance of a point from the line, squared: even under the fold,
and still not the record. -/
def depth2 (p : Pt) : Int := (p.1 - 1) * (p.1 - 1)

/-- Evenness is weaker than reading the record: the depth is fold-even
and registration changes it.
The equivalence principle asks for the stronger property. -/
theorem even_is_not_the_record :
  depth2 (fold (0, 5)) = depth2 (0, 5) ∧ depth2 (reg (0, 5)) ≠ depth2
(0, 5) := by decide

/-- The odd part of a reading under the particle map. -/
def oddPart (g : Particle → Int) (c : Particle) : Int := g c - g (bar
c)

/-- Gravity's odd part vanishes at every source: every gravitational
source stands on the line. -/
theorem every_source_on_the_line (c : Particle) : OnLine (1 + oddPart
grav c, c.energy.freq) := by
  show 1 + (c.energy.freq - c.energy.freq) = 1; omega

/-- Charge's odd part is twice the charge: every charged source stands
off the line. -/
theorem charged_source_off_the_line (c : Particle) (h : c.q3 ≠ 0) :
  ~ OnLine (1 + oddPart Particle.q3 c, c.energy.freq) := by
  show ~ (1 + (c.q3 - -c.q3) = 1); omega

/-! ## Part V. One sign: gravity cannot be screened -/

def total : List Int → Int
| [] => 0
| x :: xs => x + total xs

theorem total_nonneg (xs : List Int) (h : ∀ x ∈ xs, 0 ≤ x) : 0 ≤ total
xs := by
  induction xs with
  | nil => exact Int.le_refl 0
  | cons x xs ih =>
    have hx := h x (List.mem_cons_self)
    have hr := ih (fun y hy => h y (List.mem_cons_of_mem x hy))
    show 0 ≤ x + total xs; omega

/-- GRAVITY CANNOT BE SCREENED: sources of one sign total zero only when
every source is zero. -/
theorem one_sign_no_screening (xs : List Int) (h : ∀ x ∈ xs, 0 ≤ x) (h0 :
total xs = 0) :
  ∀ x ∈ xs, x = 0 := by
  induction xs with
  | nil => intro x hx; cases hx
  | cons y ys ih =>
    have hy := h y (List.mem_cons_self)

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have hr := total_nonneg ys (fun z hz => h z (List.mem_cons_of_mem y
hz))
have e : y + total ys = 0 := h0
have y0 : y = 0 := by omega
have t0 : total ys = 0 := by omega
intro x hx
cases hx with
| head => exact y0
| tail _ hm => exact ih (fun z hz => h z (List.mem_cons_of_mem y
hz)) t0 x hm

/-- Charge can be screened: a unit charge and its opposite total zero.
-/
theorem charge_can_be_screened : total [1, -1] = 0 ∧ (1 : Int) ≠ 0 := by
decide

/-! ## Part VI. No bit, and reversible -/

/-- One step of conservative motion under any force law f, in
Störmer-Verlet form on a pair of
successive positions. -/
def step (f : Int → Int) (s : Int × Int) : Int × Int := (s.2, 2 * s.2 -
s.1 + f s.2)
/-- Time reversal: the two positions exchanged. -/
def rev (s : Int × Int) : Int × Int := (s.2, s.1)
/-- The backward step: reverse, step, reverse. -/
def back (f : Int → Int) (s : Int × Int) : Int × Int := rev (step f
(rev s))

/-- CONSERVATIVE MOTION IS REVERSIBLE: for every force law, reversing,
stepping and reversing
undoes a step exactly. -/
theorem conservative_motion_reversible (f : Int → Int) (s : Int × Int) :
back f (step f s) = s := by
obtain (a, b) := s
show ((2 * b - (2 * b - a + f b) + f b, b) : Int × Int) = (a, b)
rw [p1]; exact (by omega, rfl)

def iter0 (g : Int × Int → Int × Int) : Nat → Int × Int → Int × Int
| 0, s => s
| n + 1, s => g (iter0 g n s)
def iterI (g : Int × Int → Int × Int) : Nat → Int × Int → Int × Int
| 0, s => s
| n + 1, s => iterI g n (g s)

/-- Any number of steps is retraced exactly by the same number of
backward steps. -/
theorem motion_retraces (f : Int → Int) (n : Nat) (s : Int × Int) :
iterI (back f) n (iter0 (step f) n s) = s := by
induction n with
| zero => rfl
| succ n ih =>
show iterI (back f) n (back f (step f (iter0 (step f) n s))) = s
rw [conservative_motion_reversible]; exact ih

/-! ## Part VII. Gravity and time co-move: redshift moves the height,
never the side -/

/-- A gravitational shift of every frequency by one factor k. -/
def redshift (k : Int) (p : Pt) : Pt := (p.1, k * p.2)

theorem redshift_equivariant (k : Int) : Equivariant (redshift k) fold
fold := fun _ => rfl
theorem redshift_keeps_the_line (k : Int) (p : Pt) : OnLine (redshift k
p) ↔ OnLine p := Iff.rfl
theorem redshift_commutates_with_registration (k : Int) (p : Pt) :
reg (redshift k p) = redshift k (reg p) := rfl

/-- Every clock co-moves: one factor shifts all heights, so every ratio
of frequencies is kept. -/
theorem clocks_co_move (k t1 t2 : Int) : t1 * (k * t2) = t2 * (k * t1)
:= by
rw [Int.mul_left_comm t1 k t2, Int.mul_left_comm t2 k t1, Int.mul_comm
t1 t2]

/-! ## Part VIII. The monic memory of the deficit -/

/-- A bound pair's gravitational reading: the parts' energies less the
binding. -/
def bound (e1 e2 b : Int) : Int := e1 + e2 - b

/-- THE DEFICIT IS READ, MONICALLY: two bindings read alike exactly when
they are equal. -/
theorem gravity_reads_the_deficit (e1 e2 b b' : Int) : bound e1 e2 b =
bound e1 e2 b' ↔ b = b' := by
unfold bound
constructor
· intro h; omega
· intro h; rw [h]

/-- A bond lowers the weight: a positive binding reads below the sum of
the parts. -/
theorem the_bond_weights_less (e1 e2 b : Int) (hb : 0 < b) : bound e1 e2
b < e1 + e2 := by
unfold bound; omega

/-! ## Part IX. The waves: two polarizations, one speed -/

/-- A gravitational wave's helicity, +2 or -2. -/
structure Wave where
freq : Int
helicity : Bool
deriving DecidableEq, Repr

def Wave.parity (w : Wave) : Wave := (w.freq, !w.helicity)

/-- The waves carry two polarizations, reversed by parity, one energy
record: the radiation carries
the one bit light carries, and the sources carry none. -/
theorem waves_two_polarizations (w : Wave) :
w.parity ≠ w ∧ w.parity.freq = w.freq ∧ w.parity.parity = w := by
obtain (f, s) := w
refine (fun e => ?_, rfl, by cases s <=> rfl)
have := congrArg Wave.helicity e
cases s <=> simp [Wave.parity] at this

/-- Ten components of a symmetric tensor, four gauge conditions, four
constraints: two polarizations. -/
theorem polarization_count : (10 - 4 - 4 : Nat) = 2 := rfl

theorem sq_pos_of_ne (m : Int) (h : m ≠ 0) : 0 < m * m := by
rcases Int.lt_or_gt_of_ne h with h' | h'
· have := Int.mul_pos (show 0 < -m by omega) (show 0 < -m by omega)
rw [Int.neg_mul_neg] at this; exact this
· exact Int.mul_pos h' h'

/-- A massive mode runs slower than a massless one; gravity's waves and
light share the cone. -/
theorem massive_slower (k ω m : Int) (h : ω * ω = k * k + m * m) (hm : m
≠ 0) : k * k < ω * ω := by
have := sq_pos_of_ne m hm
omega

/-! ## Part IX-b. Gravity reads the dark, and the vacuum's pull -/

/-- An odd reading returns zero on every state the particle map fixes.
-/
theorem odd_reading_blind (g : Particle → Int) (hodd : ∀ c, g (bar c) =
-g c) (c : Particle)
(h : bar c = c) : g c = 0 := by
have e := hodd c
rw [h] at e
omega

/-- ONLY GRAVITY READS THE DARK: a state the particle map fixes carries
no charge, every odd reading
returns zero on it, and gravity returns its energy. -/
theorem only_gravity_reads_the_dark (c : Particle) (h : bar c = c) :
Neutral c ∧ (∀ g : Particle → Int, (∀ d, g (bar d) = -g d) → g c =
0) ∧ grav c = c.energy.freq :=
{ (bar_fixed_iff_neutral c).mp h, fun g hodd => odd_reading_blind g
hodd c h, rfl }

/-- The active gravitating density ρ + 3p. -/
def active (rho p : Int) : Int := rho + 3 * p

/-- Dust attracts; the vacuum, p = -p, repels with active density -2p.
-/
theorem the_vacuum_repels (rho : Int) (h : 0 < rho) :
0 < active rho ∧ active rho (-rho) = -2 * rho ∧ active rho (-rho)
< 0 := by
unfold active; refine {?, ?, ?} <=> omega

/-- ONE SIGN OF ENERGY, NOT OF PULL: every density is positive and the
pull still reverses. The one
sign concerns the energy gravity reads, and screening would need a
negative energy. -/
theorem one_sign_of_energy_not_of_pull :
(0 : Int) < 1 ∧ active 1 (-1) < 0 ∧ total [1, 1] ≠ 0 := by decide

/-- The kinetic floor: with kinetic density K ≥ 0 and potential V, p + p
= 2K ≥ 0. -/
theorem kinetic_floor (K V : Int) (hK : 0 ≤ K) : 0 ≤ (K + V) + (K - V)
:= by omega

/-! ## Part X. The capstone -/

/-- IMMANENT GRAVITY, AT THE SEAT. Gravity reads the record and nothing
else: even under the
particle map, factoring through registration, blind to composition;
every source stands on the
line and none can be screened, while charge stands off it and can;
conservative motion is
reversible for every force law and retraces exactly; a gravitational
shift moves every height
by one factor, keeping the line, registration and every ratio of
clocks; the deficit of a bond
is read monically; the waves carry two polarizations; gravity alone
reads the dark, and the
vacuum's pressure reverses its pull. -/
theorem gravity_is_the_record :
(∀ c : Particle, grav (bar c) = grav c) ∧
(∀ p : Pt, gravPt (reg p) = gravPt p ∧ gravPt (fold p) = gravPt p) ∧
(∀ c : Particle, OnLine (1 + oddPart grav c, c.energy.freq)) ∧
(∀ xs : List Int, (∀ x ∈ xs, 0 ≤ x) → total xs = 0 → ∀ x ∈ xs, x =
0) ∧

```

```

(∀ (f : Int → Int) (n : Nat) (s : Int × Int), iterI (back f) n
(iter0 (step f) n s) = s) ∧
(∀ k : Int, Equivariant (redshift k) fold fold) ∧
(∀ k t1 t2 : Int, t1 * (k * t2) = t2 * (k * t1)) ∧
(∀ e1 e2 b b' : Int, bound e1 e2 b = bound e1 e2 b' ↔ b = b') ∧
(∀ w : Wave, w.parity ≠ w ∧ w.parity.freq = w.freq) ∧
(∀ c : Particle, bar c = c → Neutral c ∧ grav c = c.energy.freq) ∧
(∀ rho : Int, 0 < rho → active rho (-rho) < 0) :=
(gravity_even_branch, gravity_factors_through_registration,
every_source_on_the_line,
one_sign_no_screening, motion_retraces, redshift_equivariant,
clocks_co_move,
gravity_reads_the_deficit, fun w => ((waves_two_polarizations w).1,
rfl),
fun c h => ((only_gravity_reads_the_dark c h).1, rfl), fun rho h =>
(the_vacuum_repels rho h).2.2)

```

end SPHYS.Gravity

```

#print axioms SPHYS.Gravity.pe
#print axioms SPHYS.Gravity.fold_involutive
#print axioms SPHYS.Gravity.seat_fixed_line
#print axioms SPHYS.Gravity.reg_lands
#print axioms SPHYS.Gravity.reg_fixes_iff
#print axioms SPHYS.Gravity.reg_forgets_side
#print axioms SPHYS.Gravity.least_erasure_iff_value
#print axioms SPHYS.Gravity.equivariant_id
#print axioms SPHYS.Gravity.equivariant_comp
#print axioms SPHYS.Gravity.fix_functorial
#print axioms SPHYS.Gravity.fold_global_seat
#print axioms SPHYS.Gravity.equivariant_carrier_lands
#print axioms SPHYS.Gravity.value_on_image
#print axioms SPHYS.Gravity.kinetic_crossing
#print axioms SPHYS.Gravity.off_locus_pair
#print axioms SPHYS.Gravity.ee
#print axioms SPHYS.Gravity.mirror_involutive
#print axioms SPHYS.Gravity.psi_phi
#print axioms SPHYS.Gravity.phi_psi
#print axioms SPHYS.Gravity.phi_injective
#print axioms SPHYS.Gravity.phi_equivariant
#print axioms SPHYS.Gravity.mirror_fixed_iff_real
#print axioms SPHYS.Gravity.phi_line_iff_real
#print axioms SPHYS.Gravity.stationary_iff_real
#print axioms SPHYS.Gravity.line_is_stationary

```

```

'SPHYS.Gravity.pe' does not depend on any axioms
'SPHYS.Gravity.fold_involutive' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.seat_fixed_line' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.reg_lands' does not depend on any axioms
'SPHYS.Gravity.reg_fixes_iff' depends on axioms: [propext]
'SPHYS.Gravity.reg_forgets_side' does not depend on any axioms
'SPHYS.Gravity.least_erasure_iff_value' depends on axioms: [propext]
'SPHYS.Gravity.equivariant_id' does not depend on any axioms
'SPHYS.Gravity.equivariant_comp' does not depend on any axioms
'SPHYS.Gravity.fix_functorial' does not depend on any axioms
'SPHYS.Gravity.fold_global_seat' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.equivariant_carrier_lands' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.value_on_image' does not depend on any axioms
'SPHYS.Gravity.kinetic_crossing' does not depend on any axioms
'SPHYS.Gravity.off_locus_pair' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.ee' does not depend on any axioms
'SPHYS.Gravity.mirror_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.psi_phi' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.phi_psi' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.phi_injective' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.phi_equivariant' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.mirror_fixed_iff_real' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.phi_line_iff_real' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.stationary_iff_real' depends on axioms: [propext]
'SPHYS.Gravity.line_is_stationary' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.measured_on_line' does not depend on any axioms
'SPHYS.Gravity.no_measurement_off_line' does not depend on any axioms
'SPHYS.Gravity.energy_carrier_lands' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.bar_involutive' does not depend on any axioms
'SPHYS.Gravity.bar_fixed_iff_neutral' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.odd_charges_even_energy' does not depend on any axioms
'SPHYS.Gravity.gravity_reads_no_charge_bit' does not depend on any
axioms
'SPHYS.Gravity.involutions_commute' does not depend on any axioms
'SPHYS.Gravity.tmirror_involutive' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.joint_fixed_iff' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.charge_carrier_equivariant' depends on axioms: [propext,
Quot.sound]

```

```

#print axioms SPHYS.Gravity.measured_on_line
#print axioms SPHYS.Gravity.no_measurement_off_line
#print axioms SPHYS.Gravity.energy_carrier_lands
#print axioms SPHYS.Gravity.bar_involutive
#print axioms SPHYS.Gravity.bar_fixed_iff_neutral
#print axioms SPHYS.Gravity.odd_charges_even_energy
#print axioms SPHYS.Gravity.gravity_reads_no_charge_bit
#print axioms SPHYS.Gravity.involutions_commute
#print axioms SPHYS.Gravity.tmirror_involutive
#print axioms SPHYS.Gravity.joint_fixed_iff
#print axioms SPHYS.Gravity.charge_carrier_equivariant
#print axioms SPHYS.Gravity.pair_lands_once
#print axioms SPHYS.Gravity.tmirror_equivariant
#print axioms SPHYS.Gravity.neutral_charge_on_line
#print axioms SPHYS.Gravity.pair_on_the_edges
#print axioms SPHYS.Gravity.stable_lands
#print axioms SPHYS.Gravity.gravity_even_branch
#print axioms SPHYS.Gravity.gravity_factors_through_registration
#print axioms SPHYS.Gravity.composition_blind
#print axioms SPHYS.Gravity.even_is_not_the_record
#print axioms SPHYS.Gravity.every_source_on_the_line
#print axioms SPHYS.Gravity.charged_source_off_the_line
#print axioms SPHYS.Gravity.total_nonneg
#print axioms SPHYS.Gravity.one_sign_no_screening
#print axioms SPHYS.Gravity.charge_can_be_screened
#print axioms SPHYS.Gravity.conservative_motion_reversible
#print axioms SPHYS.Gravity.motion_retraces
#print axioms SPHYS.Gravity.redshift_equivariant
#print axioms SPHYS.Gravity.redshift_keeps_the_line
#print axioms SPHYS.Gravity.redshift_commutes_with_registration
#print axioms SPHYS.Gravity.clocks_co_move
#print axioms SPHYS.Gravity.gravity_reads_the_deficit
#print axioms SPHYS.Gravity.the_bond_weights_less
#print axioms SPHYS.Gravity.waves_two_polarizations
#print axioms SPHYS.Gravity.polarization_count
#print axioms SPHYS.Gravity.sq_pos_of_ne
#print axioms SPHYS.Gravity.massive_slower
#print axioms SPHYS.Gravity.odd_reading_blind
#print axioms SPHYS.Gravity.only_gravity_reads_the_dark
#print axioms SPHYS.Gravity.the_vacuum_repels
#print axioms SPHYS.Gravity.one_sign_of_energy_not_of_pull
#print axioms SPHYS.Gravity.kinetic_floor
#print axioms SPHYS.Gravity.gravity_is_the_record

```

```

'SPHYS.Gravity.pair_lands_once' does not depend on any axioms
'SPHYS.Gravity.tmirror_equivariant' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.neutral_charge_on_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.pair_on_the_edges' does not depend on any axioms
'SPHYS.Gravity.stable_lands' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.gravity_even_branch' does not depend on any axioms
'SPHYS.Gravity.gravity_factors_through_registration' does not depend on
any axioms
'SPHYS.Gravity.composition_blind' does not depend on any axioms
'SPHYS.Gravity.even_is_not_the_record' does not depend on any axioms
'SPHYS.Gravity.every_source_on_the_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.charged_source_off_the_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.total_nonneg' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.one_sign_no_screening' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.charge_can_be_screened' does not depend on any axioms
'SPHYS.Gravity.conservative_motion_reversible' depends on axioms:
[propext, Quot.sound]
'SPHYS.Gravity.motion_retraces' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.redshift_equivariant' does not depend on any axioms
'SPHYS.Gravity.redshift_keeps_the_line' does not depend on any axioms
'SPHYS.Gravity.redshift_commutes_with_registration' does not depend on
any axioms
'SPHYS.Gravity.clocks_co_move' depends on axioms: [propext]
'SPHYS.Gravity.gravity_reads_the_deficit' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.the_bond_weights_less' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.waves_two_polarizations' depends on axioms: [propext]
'SPHYS.Gravity.polarization_count' does not depend on any axioms
'SPHYS.Gravity.sq_pos_of_ne' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.massive_slower' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.odd_reading_blind' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.only_gravity_reads_the_dark' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.the_vacuum_repels' depends on axioms: [propext,
Quot.sound]
'SPHYS.Gravity.one_sign_of_energy_not_of_pull' does not depend on any
axioms
'SPHYS.Gravity.kinetic_floor' depends on axioms: [propext, Quot.sound]
'SPHYS.Gravity.gravity_is_the_record' depends on axioms: [propext,
Quot.sound]

```

APPENDIX B: GRAVITY_TWIN.F90 AND ITS RUN

```
! Gravity_Twin.f90 · the executed twin of SPHYS_Gravity.lean · Fortran
2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Gravity_Twin.f90
! Blocks: A gravity reads the record; B the equivalence principle and
the deficit; C one sign, no
! screening; D reversible motion; E gravity and time co-move; F the
classical tests computed from
! the constants; G the waves; H the two branches' prices; I the sky's
budget: the vacuum's pull.
program gravity_twin
implicit none
integer, parameter :: dp = selected_real_kind(15, 307)
integer, parameter :: i8 = selected_int_kind(18)
real(dp), parameter :: CL = 299792458.0_dp, GMS = 1.32712440018e20_dp,
KB = 1.380649e-23_dp
real(dp), parameter :: PI = 3.141592653589793_dp, ARCSEC =
206264.80624709636_dp
integer :: checks, fails
integer(i8) :: seed
checks = 0; fails = 0; seed = 20260930_i8
call block_a(); call block_b(); call block_c(); call block_d()
call block_e(); call block_g(); call block_h(); call
block_i()
write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
if (fails > 0) error stop 1
contains
subroutine check(name, ok)
character(*), intent(in) :: name
logical, intent(in) :: ok
checks = checks + 1
if (.not. ok) fails = fails + 1
write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name
end subroutine check

function rnd() result(r)
real(dp) :: r
seed = modulo(16807_i8*seed, 2147483647_i8)
r = real(seed, dp)/2147483647.0_dp
end function rnd

subroutine block_a()
integer :: q, col, e, n, neven, nblind, nsrc, noff
write(*, '(a)') 'A · gravity reads the record: even under the
particle map, blind to the side'
n = 0; neven = 0; nblind = 0; nsrc = 0; noff = 0
do q = -6, 6
do col = -2, 2
do e = -20, 20
n = n + 1
! the particle (q, col, e) and its antiparticle (-q, -col, e):
gravity reads e alone
if (grav(q, e) == grav(-q, e)) neven = neven + 1
if (grav(q, e) == grav(0, e)) nblind = nblind + 1
if (1 + (grav(q, e) - grav(-q, e)) == 1) nsrc = nsrc + 1
if (q /= 0 .and. 1 + (q - (-q)) /= 1) noff = noff + 1
end do
end do
call check('gravity reads a particle and its antiparticle (-q,
-colour, e) alike (2665 of 2665)', neven == n)
call check('gravity is blind to charge at fixed energy (2665 of
2665)', nblind == n)
call check('every gravitational source stands on the line; every
charged source off it', &
nsrc == n .and. noff == n - 5*41)
end subroutine block_a

pure function grav(q, e) result(g)
integer, intent(in) :: q, e
integer :: g
g = e + 0*q
end function grav

subroutine block_b()
real(dp) :: bti, bpt, mu, fti, fpt, dfrac, eta, reach
write(*, '(a)') 'B · the equivalence principle, and the deficit read
with the rest'
bti = 8.72301_dp; bpt = 7.92655_dp; mu = 931.49410242_dp
fti = bti/mu; fpt = bpt/mu; dfrac = fti - fpt
eta = 1.5e-15_dp + sqrt(2.3e-15_dp**2 + 1.5e-15_dp**2)
reach = eta/dfrac
write(*, '(a,f8.5,a,f8.5,a,es10.3)') ' binding fraction of the mass:
titanium ', 100.0_dp*fti, ' %, platinum ', &
100.0_dp*fpt, ' %, difference ', dfrac
write(*, '(a,es10.3,a,es10.3)') ' MICROSCOPE [eta] <= ', eta, ': the
binding deficit falls with the rest to ', reach
call check('the deficit of nuclear binding is read by gravity to
better than 1e-11 of itself', reach < 1.0e-11_dp)
call check('antihydrogen falls down, a = (0.75 +/- 0.21) g, repulsion
excluded (ALPHA-g 2023)', &
0.75_dp - 3.0_dp*sqrt(0.13_dp**2 + 0.16_dp**2) > -1.0_dp)
```

```
end subroutine block_b

subroutine block_c()
integer :: i, j, nq, nm, s, qq(4), mm(4)
write(*, '(a)') 'C · one sign: gravity cannot be screened, charge
can'
nq = 0; nm = 0
do i = 1, 20000
s = 0
do j = 1, 4
qq(j) = int(7.0_dp*rnd()) - 3
if (qq(j) == 0) qq(j) = 1
end do
if (sum(qq) == 0) nq = nq + 1
do j = 1, 4
mm(j) = int(4.0_dp*rnd())
end do
if (maxval(mm) > 0 .and. sum(mm) == 0) nm = nm + 1
end do
write(*, '(a,i0,a,i0)') ' of 20000 random sets of four nonzero
charges, neutral: ', nq, &
'; of 20000 non-empty mass sets, massless: ', nm
call check('charges screen (neutral sets exist); masses never total
zero unless all are zero', nq > 0 .and. nm == 0)
end subroutine block_c

subroutine block_d()
integer :: n, nsteps
real(dp) :: dt, x0(2), x1(2), x2(2), a(2), e0, e, emax, v(2), xa(2),
xb(2), err
write(*, '(a)') 'D · no bit, and reversible: a Kepler orbit run
forward and backward'
nsteps = 50000; dt = 10.0_dp*2.0_dp*PI/real(nsteps, dp)
! GM = 1, a = 1, eccentricity 1/2, start at perihelion
x0 = [0.5_dp, 0.0_dp]; v = [0.0_dp, sqrt(3.0_dp)]
a = acc(x0); x1 = x0 + dt*v + 0.5_dp*dt*dt*a
e0 = 0.5_dp*sum(v*v) - 1.0_dp/sqrt(sum(x0*x0)); emax = 0.0_dp
xa = x0; xb = x1
do n = 1, nsteps
x2 = 2.0_dp*xb - xa + dt*dt*acc(xb)
v = (x2 - xa)/(2.0_dp*dt)
e = 0.5_dp*sum(v*v) - 1.0_dp/sqrt(sum(xb*xb))
emax = max(emax, abs(e/e0 - 1.0_dp))
xa = xb; xb = x2
end do
! reverse: exchange the last two positions and run the same number
of steps
x2 = xa; xa = xb; xb = x2
do n = 1, nsteps
x2 = 2.0_dp*xb - xa + dt*dt*acc(xb)
xa = xb; xb = x2
end do
err = sqrt(sum((xb - x0)**2))
write(*, '(a,es9.2,a,es9.2)') ' ten revolutions: max |dE/E| = ',
emax, '; retrace error after reversal = ', err
call check('the orbit keeps its energy to 1e-5 over ten
revolutions', emax < 1.0e-5_dp)
call check('run backwards, the orbit retraces its start to 1e-9',
err < 1.0e-9_dp)
end subroutine block_d

pure function acc(x) result(g)
real(dp), intent(in) :: x(2)
real(dp) :: g(2), r
r = sqrt(sum(x*x))
g = -x/(r*r)
end function acc

subroutine block_e()
integer :: i
real(dp) :: f1, f2, phi, k, worst, pr, clk
write(*, '(a)') 'E · gravity and time co-move: one factor shifts
every clock'
worst = 0.0_dp
do i = 1, 10000
f1 = 10.0_dp**(9.0_dp + 6.0_dp*rnd()); f2 = 10.0_dp**(9.0_dp +
6.0_dp*rnd())
phi = -1.0e-8_dp*rnd()
k = 1.0_dp + phi
worst = max(worst, abs((k*f1)/(k*f2)/(f1/f2) - 1.0_dp))
end do
write(*, '(a,es9.2)') ' 10000 clock pairs in 10000 potentials: max
change of their ratio ', worst
call check('every ratio of clocks is kept by the gravitational shift
(to 1e-15)', worst < 1.0e-15_dp)
pr = 9.80665_dp*22.5_dp/CL**2; clk = 9.80665_dp*0.33_dp/CL**2
write(*, '(a,es10.3,a,es10.3)') ' Pound-Rebka tower 22.5 m: dnu/nu =
', pr, '; optical clocks 33 cm: ', clk
call check('the tower shift is 2.455e-15 and the 33 cm shift
3.60e-17', &
abs(pr/2.4550e-15_dp - 1.0_dp) < 1.0e-3_dp .and.
abs(clk/3.6007e-17_dp - 1.0_dp) < 1.0e-3_dp)
end subroutine block_e
```



```

subroutine block_f()
  real(dp) :: defl, prec, amerc, emerc, porb, pdot, pb, ecc, m1, m2,
  tsun, fe
  write(*,'(a)') 'F · the classical floor, computed from the
  constants'
  defl = 4.0_dp/GMS/(CL**2*6.957e8_dp)*ARCSEC
  amerc = 5.7909050e10_dp; emerc = 0.205630_dp; porb = 87.9691_dp
  prec = 6.0_dp*PI*GMS/(CL**2*amerc*(1.0_dp -
  emerc**2))*(36525.0_dp/porb)*ARCSEC
  tsun = GMS/CL**3
  pb = 0.322997448918_dp*86400.0_dp; ecc = 0.6171340_dp; m1 =
  1.438_dp; m2 = 1.390_dp
  fe = (1.0_dp + 73.0_dp/24.0_dp*ecc**2 +
  37.0_dp/96.0_dp*ecc**4)/(1.0_dp - ecc**2)**3.5_dp
  pdot = -192.0_dp*PI/5.0_dp*tsun**((5.0_dp/3.0_dp)*(pb/(2.0_dp*PI))**
  (-5.0_dp/3.0_dp)*fe*m1*m2/(m1 +
  m2)**(1.0_dp/3.0_dp)
  write(*,'(a,f7.4,a,f7.3,a,es12.5)') ' deflection at the solar limb
  ', defl, ' arcsec; Mercury ', prec, &
  ' arcsec per century; PSR B1913+16 dPb/dt = ', pdot
  call check('light grazing the Sun bends 1.751 arcsec, twice the
  Newtonian angle', abs(defl - 1.751_dp) < 2.0e-3_dp)
  call check('Mercury's perihelion advances 42.98 arcsec per
  century', abs(prec - 42.98_dp) < 0.05_dp)
  call check('the binary pulsar's orbital decay is -2.4026e-12, the
  GR value (Weisberg and Huang 2016)', &
  abs(pdot/(-2.40263e-12_dp) - 1.0_dp) < 2.0e-3_dp)
end subroutine block_f

subroutine block_g()
  integer :: s, n
  real(dp) :: dv
  write(*,'(a)') 'G · the waves: two polarizations, the speed of
  light'
  n = 0
  do s = -1, 1, 2
    if (-s /= s) n = n + 1
  end do
  dv = 1.74_dp/(130.0e6_dp*3.15576e7_dp)

```

```

  write(*,'(a,i0,a,es9.2)') ' polarizations 10 - 4 - 4 = ', 10 - 4 -
  4, ' ; GW170817 dv/c <= ', dv
  call check('two polarizations, each moved by parity, and gravity at
  the speed of light to 1e-15', &
  n == 2 .and. 10 - 4 - 4 == 2 .and. dv < 1.0e-15_dp)
end subroutine block_g

subroutine block_h()
  real(dp) :: floor
  write(*,'(a)') 'H · the two branches: the reversible run registers
  nothing; the kept bit is priced'
  floor = KB*300.0_dp*log(2.0_dp)
  write(*,'(a,f6.1,a)') ' gravity's reversed orbit: 0 bits
  committed; the magnet's bit: ', 5.0e-19_dp/floor, &
  ' floors per switch'
  call check('the reversible branch commits no bit while the priced
  branch pays above the floor', &
  5.0e-19_dp/floor > 100.0_dp)
end subroutine block_h

subroutine block_i()
  real(dp) :: om, ol, q0, qm, zacc, zeq
  write(*,'(a)') 'I · the sky's budget: dust attracts, the vacuum
  repels, and the sign is read'
  om = 0.3153_dp; ol = 0.6847_dp
  q0 = om/2.0_dp - ol; qm = 1.0_dp/2.0_dp
  zacc = (2.0_dp*ol/om)**(1.0_dp/3.0_dp) - 1.0_dp
  zeq = (ol/om)**(1.0_dp/3.0_dp) - 1.0_dp
  write(*,'(a,f7.4,a,f5.2,a,f6.3,a,f6.3)') ' flat budget of Planck
  2018: q0 = ', q0, ' (all matter: ', qm, &
  '); acceleration begins at z = ', zacc, ' ; the vacuum
  overtakes matter at z = ', zeq
  call check('the two budgets give decelerations of opposite sign: q0
  = -0.527 against +0.5', &
  abs(q0 + 0.527_dp) < 1.0e-3_dp .and. qm > 0.0_dp)
  call check('the acceleration begins at z = 0.63 and the vacuum
  overtakes matter at z = 0.295', &
  abs(zacc - 0.631_dp) < 2.0e-3_dp .and. abs(zeq -
  0.295_dp) < 2.0e-3_dp)
end subroutine block_i
end program gravity_twin

```

A · gravity reads the record: even under the particle map, blind to the side

PASS gravity reads a particle and its antiparticle (-q, -colour, e) alike (2665 of 2665)

PASS gravity is blind to charge at fixed energy (2665 of 2665)

PASS every gravitational source stands on the line; every charged source off it

B · the equivalence principle, and the deficit read with the rest

binding fraction of the mass: titanium 0.93645 %, platinum 0.85095 %, difference 8.550E-04

MICROSCOPE [eta] <= 4.246E-15: the binding deficit falls with the rest to 4.966E-12

PASS the deficit of nuclear binding is read by gravity to better than 1e-11 of itself

PASS antihydrogen falls down, a = (0.75 +- 0.21) g, repulsion excluded (ALPHA-g 2023)

C · one sign: gravity cannot be screened, charge can

of 20000 random sets of four nonzero charges, neutral: 1834; of 20000 non-empty mass sets, massless: 0

PASS charges screen (neutral sets exist); masses never total zero unless all are zero

D · no bit, and reversible: a Kepler orbit run forward and backward

ten revolutions: max |dE/E| = 4.29E-06; retrace error after reversal = 5.17E-10

PASS the orbit keeps its energy to 1e-5 over ten revolutions

PASS run backwards, the orbit retraces its start to 1e-9

E · gravity and time co-move: one factor shifts every clock

10000 clock pairs in 10000 potentials: max change of their ratio 3.33E-16

PASS every ratio of clocks is kept by the gravitational shift (to 1e-15)

Pound-Rebka tower 22.5 m: dnu/nu = 2.455E-15; optical clocks 33 cm: 3.601E-17

PASS the tower shift is 2.455e-15 and the 33 cm shift 3.60e-17

F · the classical floor, computed from the constants

deflection at the solar limb 1.7512 arcsec; Mercury 42.981 arcsec per century; PSR B1913+16 dPb/dt = -2.40210E-12

PASS light grazing the Sun bends 1.751 arcsec, twice the Newtonian angle

PASS Mercury's perihelion advances 42.98 arcsec per century

PASS the binary pulsar's orbital decay is -2.4026e-12, the GR value (Weisberg and Huang 2016)

G · the waves: two polarizations, the speed of light

polarizations 10 - 4 - 4 = 2; GW170817 dv/c <= 4.24E-16

PASS two polarizations, each moved by parity, and gravity at the speed of light to 1e-15

H · the two branches: the reversible run registers nothing; the kept bit is priced

gravity's reversed orbit: 0 bits committed; the magnet's bit: 174.2 floors per switch

PASS the reversible branch commits no bit while the priced branch pays above the floor

I · the sky's budget: dust attracts, the vacuum repels, and the sign is read

flat budget of Planck 2018: q0 = -0.5271 (all matter: 0.50); acceleration begins at z = 0.632; the vacuum overtakes matter at z = 0.295

PASS the two budgets give decelerations of opposite sign: q0 = -0.527 against +0.5

PASS the acceleration begins at z = 0.63 and the vacuum overtakes matter at z = 0.295

BATTERY-JSON: {"checks":17,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

PhysOS^T 1.0.5p · RECEIPT · mode quick

TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0

MANIFEST sha256 a4108cd26bcbce6f, the files that ran

GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held by review, clean

EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 · registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0

REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}

WITNESS 5 of 5 twins ran, each held to one battery line with zero failures

CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b

ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)

SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0

laws: 68

JUDGMENT refused 68 of 68 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True